Revise change of phase:
Solid to liquid = melting
Liquid to gas = evaporation
Liquid to solid = freezing or solidification
Gas to liquid = condensation
Solid to gas or gas to solid = sublimation

Observing evaporation

Aim: to observe the types of crystals that grow at different rates of evaporation.
Materials:
Alum solution
Bunsen burner
Tripod and gauze mat
2 petri dishes
evaporating basin
heatproof mat
Method:

1. Divide the alum solution between the evaporating dish and the two petri dishes.
2. Place the evaporating dish on the tripod and gauze mat and heat gently using the Bunsen burner. (Do not allow the solution to boil rapidly as it may start spitting.
3. When all the water has evaporated, observe the size of the crystals that have formed.
4. Place on petri dish in a warm position.
5. Place the second petri dish in the refrigerator.
6. After three days observe the size of the crystals in the two petri dishes.
7. Record your observations in a table with the headings ‘rapid crystallisation’, ‘slow crystallisation’, ‘very slow crystallisation’.
8. Write a conclusion for your results.

Revise particle model:
All substances are made up of tiny particles which are constantly vibrating.
Heating speeds up the movement of these particles while cooling slows it down.
If movement is speeded up particles move more rapidly and are not held together as closely as before.

Diffusion – the ‘spreading out’ of a substance. A substance diffuses from an area where there is more of it to an area where there is less.
Eg. When 2 gases are placed in a container together, the particles of each gas will spread or diffuse through each other until they are evenly spread out in the container.

Diffusion in a liquid
Aim: To observe what happens when a crystal of potassium permanganate is allowed to stand in a beaker of water.
Materials:
250mL beaker
spatula

crystal of potassium permanganate
Method:

1. Three-quarters fill a beaker of water and wait for it to be still.
2. Carefully lower a crystal of potassium permanganate down the side of the beaker.
3. Observe what happens immediately, after 20 minutes and after one day.
4. Draw a series of diagrams to record your observations.
5. In your conclusion, explain what has happened in terms of particle movement.

Diffusion of a gas
Release a small amount of perfume in a corner of the room. Have students sitting at different points of the room with their eyes closed, raise their hands as soon as they can smell the odour. Two students map the response to show how the odour travels around the room.

Questions
Why has the odour in this experiment travelled across the room?
Where is the odour the strongest: near the source or on the other side of the room? Why?